Tree Builder

Working out what a cladogram claims, what it does not claim, and how counting changes decides between two arrangements that both look reasonable.

Name ______________ Date ______________ Period ______________

Demonstration: andresdbp.github.io/resources/demos/tree-builder/

1 Before you open anything

Answer from what you already know. Having something on paper is the point; being right at this stage is not.

PREDICTION

A cladogram is drawn with lizard on one line and pigeon on the line below it, joined at a single branch point. Someone redraws the same diagram with pigeon on top and lizard below — the branch point is spun through 180° and nothing else is touched.

a. Does the redrawn diagram make a different claim about which animals share which ancestors? Yes or no.

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b. The demonstration scores a tree by the number of evolutionary changes it has to assume. After the redraw, does that number go up, go down, or stay the same?

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Write one sentence saying how you decided.

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2 Build one tree, and watch the score fall

Open the demonstration. Panel **1 · Build the tree** holds one chip per animal. Clicking two chips joins them into a clade. The lancelet chip is greyed out — it is the outgroup and stays at the root.

1. Before joining anything, record the top row of the table from panel **2 · The score**.
2. Click **Lizard**, then **Pigeon**. Read the note that appears under the chips: it names the shared derived characters supporting the join. Record how many there are, then fill in the readouts.
3. Click **Human**, then **Lizard + Pigeon**.

4. Click **Salamander**, then **Human + Lizard + Pigeon**.
5. Click **Shark**, then the one remaining chip.

After this join	Shared derived characters named	Steps (changes)	Groups left to join	Rank of all 105 trees
nothing joined yet				
Lizard + Pigeon				
Human + Lizard + Pigeon				
Salamander + Human + Lizard + Pigeon				
Shark + Salamander + Human + Lizard + Pigeon				

> **Rank of all 105 trees** reads as a dash until every animal is joined. A half-built picture is not yet one of the 105 arrangements, so there is nothing to rank it against.

a. The final join added one shared derived character but did not lower the step count at all. Explain why joining the last two remaining groups cannot change the score.

b. Copy the tree string from the **Your tree** label above the diagram, exactly as written.

3 What a rotation costs

Keep the tree you just built. A node is a hinge: turning it re-orders what is drawn above and below. This part decides whether your page-1 prediction holds.

1. Record row 0 of the table before touching anything.
2. On the diagram, click the coloured circle at the point where **lizard** and **pigeon** join. Record row 1.
3. Press **Rotate a node** four more times, recording a row each time. The button works through a different hinge on each press, so the picture keeps changing.

Rotations you made	The string in the *Your tree* label	Steps (changes)	Rank of all 105 trees
0			
1			
2			
3			
4			
5			

a. One column changed on almost every row and two columns never changed at all. State the rule in your own words: what does a rotation alter, and what does it leave alone?

b. Go back to your page-1 prediction. Was each part right? If either was not, name the specific assumption that turned out to be false.

c. On at least one of your rows, two animals ended up printed on neighbouring lines that were not neighbours before. Using your rule, explain why "printed next to each other" is not a statement about relatedness.

d. If a rotation does not change the information in a tree, then what exactly *is* the information a cladogram carries? Answer in one sentence.

4 When a good-looking grouping still loses

Human and pigeon share two striking derived features that nothing else on this tree has. Grouping them is a defensible move. This part prices it.

1. Press **Start over**, then tick **Show how many shared derived characters each join would have**.
2. Click **Human**. Each other chip now carries a count. Record them, then click **Human** again to deselect.
3. Click **Lizard** and record the same counts.

Chip you selected	Shark	Salamander	Human	Lizard	Pigeon
Human			—		
Lizard				—	

1. Now build the human-and-pigeon tree, clicking the new animal first each time: **Human** then **Pigeon**; **Lizard** then the **Human + Pigeon** chip; **Salamander** then that chip; **Shark** then the last chip. Record as you go.
2. Press **Cold-blooded together** in panel **3 · Try another arrangement** and record its score.

Arrangement	Steps (changes)	Rank of all 105 trees
Human + Pigeon		—
Lizard + Human + Pigeon		—
Salamander + Lizard + Human + Pigeon		—
complete tree, human and pigeon grouped		
Cold-blooded together		
your part-2 tree, for comparison		

a. Grouping human with pigeon is supported by real shared derived characters, and it still scores worse than your part-2 tree. Using the two counts you recorded above, account for the difference exactly.

b. Count the entries in the list of scored trees in panel 3. Now press **The ladder**, read its score, and count the entries again. The list did not grow. Explain what that tells you about *The ladder* and the tree you just built by hand.

c. *Cold-blooded together* groups shark, salamander and lizard on a feature all three genuinely share: none of them holds a constant internally set body temperature. Explain why that shared feature is not evidence that the three form a clade.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Then reopen it and check yourself — and where you were wrong, write down what you had assumed.

The whole character matrix, condensed. *Derived* means the state differs from the lancelet's.

Character	Derived in
1 — notochord at some stage	no animal — all six have it
2 — vertebral column	shark, salamander, human, lizard, pigeon
3, 4, 5 — bony skeleton; lungs; four limbs with digits	salamander, human, lizard, pigeon
6, 7 — amniotic egg; waterproof keratinised skin	human, lizard, pigeon
8, 9, 10 — two openings behind the eye socket; uric acid waste; beta-keratin foot scales	lizard, pigeon
11, 12 — constant internally set body temperature; four-chambered heart	human, pigeon
13 — feathers	pigeon
14 — hair	human

1. Two trees differ in one respect only. In tree A, lizard and pigeon are sister taxa and human branches just outside them. In tree B, human and pigeon are sister taxa and lizard branches just outside them. Everything else — lancelet at the root, then shark, then salamander — is identical.

Work out the number of steps characters 8, 9, 10, 11 and 12 each need on tree A and on tree B, total the five for each tree, and state which whole tree scores lower and by how much.

2. Character 2, the vertebral column, is derived in all five animals other than the lancelet. State how many steps it costs on every one of the 105 possible trees, and say whether it can be used to choose between any two of them. Justify the number.

3. Someone argues that feathers and hair are such conspicuous, wholly different structures that they are strong evidence pigeon and human are *not* close relatives. Explain what is wrong with the argument. No calculation.

4. The lancelet is never joined to anything and never appears in a clade you build. Name the two separate jobs it does in this analysis, and state what you would no longer be able to determine if it were removed from the matrix.

6 On your own

Nothing here is graded. The demonstration does considerably more than this worksheet has used.

- Predict the score of **Swimmers together** before you press it, then use the *Steps* column of the character matrix to find which characters that arrangement forces to change more than once.
- Tick **Show only the parsimony-informative characters** and count how many of the fourteen survive. The ones that drop out are real biology. Say precisely what they can and cannot tell you.
- Tap a single row of the character matrix to trace that one character on the tree; the tick marks then read *gained* or *lost*. Find a character costing two steps and decide whether the demonstration is drawing it as two separate gains or as a gain followed by a loss — then ask whether counting steps alone could ever distinguish the two.